



Desarrollo de Sistemas Móviles

Tools: Laboratorio 2026

Al finalizar la clase, serás capaz de:

- Definir qué es una aplicación móvil y sus tipos principales
- Describir la evolución del ecosistema móvil hasta el 2026
- Distinguir las arquitecturas de Android e iOS
- Identificar el stack tecnológico del curso: Kotlin + Jetpack Compose
- Dominar el Stack 2026: Kotlin, Compose, **SpringBoot** o **FastAPI** y AI on-device.



....

Comenzamos 8:25 am ...

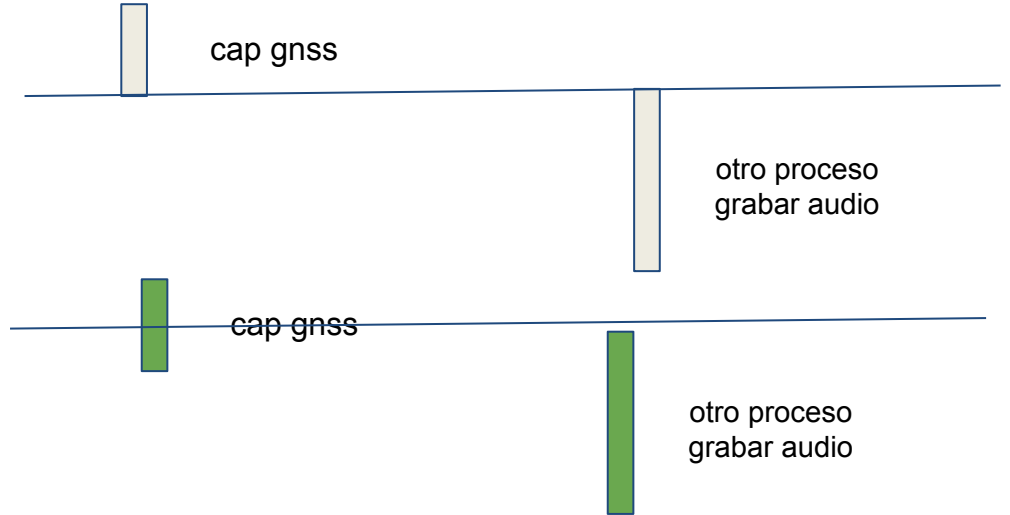
Hoy habilitaremos llaves y credenciales para trabajar con algunos servicio, existentes pero necesarios: DB, Github actions, blob repository (S3/R2), auth google

Suspend

thread 1



thread 2



— App —————

APP_NAME=Mobile API
APP_ENV=development
DEBUG=true
LOG_LEVEL=INFO

— PostgreSQL —————

POSTGRES_HOST=localhost
POSTGRES_PORT=5432
POSTGRES_DB=mobile_db
POSTGRES_USER=postgres
POSTGRES_PASSWORD=postgres

— MongoDB —————

MONGO_URI=mongodb+srv://mongo_user:mongo_password@kankunapaq-db.me4ilfe.mongodb.net/
MONGO_DATABASE=test

— Redis —————

REDIS_URL=redis://localhost:6379

— Object Storage (Cloudflare R2 / AWS S3) —————

OBJECT_STORAGE_BUCKET=mobile
OBJECT_STORAGE_ENDPOINT=https://your-account-id.r2.cloudflarestorage.com
OBJECT_STORAGE_ACCESS_KEY=your_access_key
OBJECT_STORAGE_SECRET_KEY=your_secret_key
OBJECT_STORAGE_REGION=auto

— CORS —————

ALLOWED_ORIGINS=http://localhost:3000,http://localhost:5173,https://mobile_endpoints.kankunapaq.com

— Authentication (JWT) —————

SECRET_KEY=change_this_to_a_long_random_secret_key
ALGORITHM=HS256
ACCESS_TOKEN_EXPIRE_MINUTES=120
REFRESH_TOKEN_EXPIRE_DAYS=30

— Google Login —————

GOOGLE_CLIENT_ID=your-google-client-id.apps.googleusercontent.com
GOOGLE_CLIENT_SECRET=your-google-client-secret
GOOGLE_REDIRECT_URI=http://localhost:3000/auth/google/callback

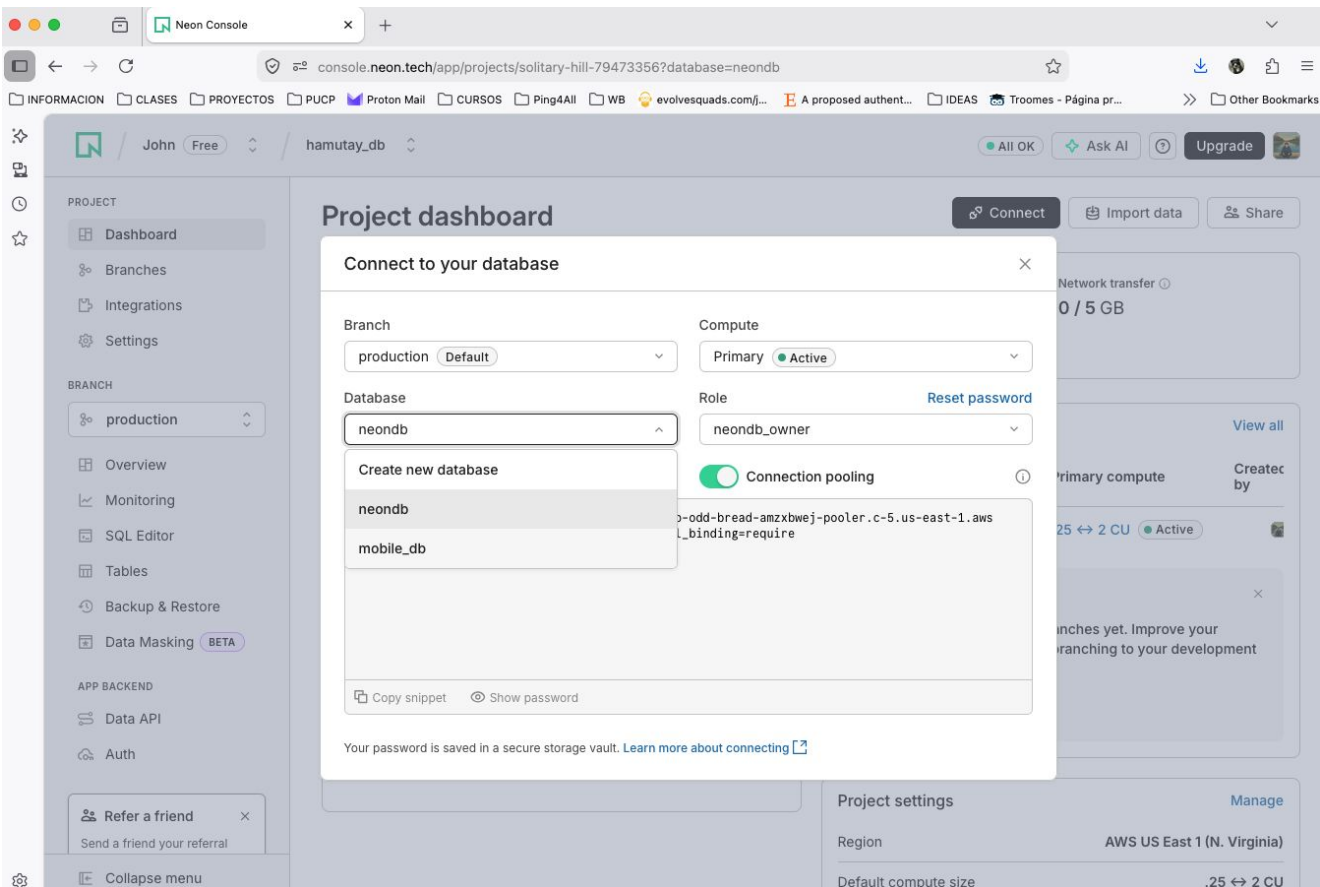
— Seed —————

SEED_ADMIN_EMAIL=admin@mobileapi.com
SEED_ADMIN_PASSWORD=Admin1234! # PBKDF2WithHmacSHA256
SEED_ADMIN_DOCUMENT_ID=00000000

— LLM —————

LLM_TYPE="local" # local or cloud
LLM_URL=http://127.0.0.1:9000

LLM_PROVIDER=openai
LLM_BASE_URL=https://api.openai.com/v1
LLM_API_KEY=xxx



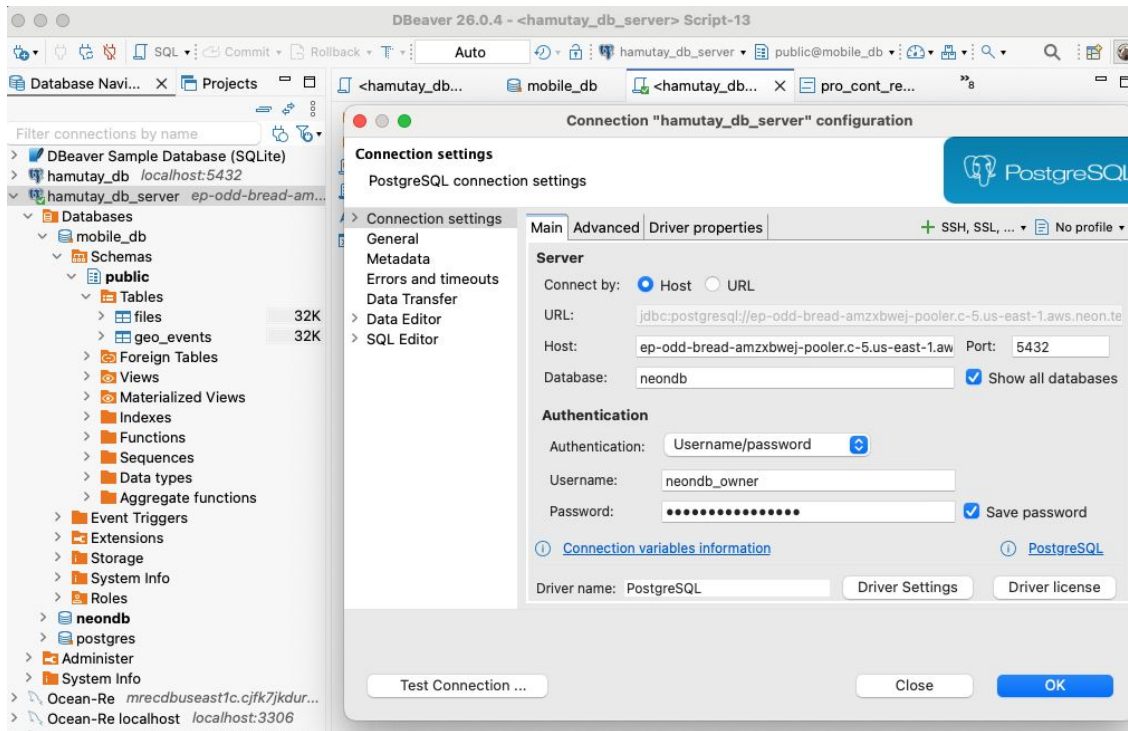
The screenshot shows the Neon Console web interface. A modal titled "Connect to your database" is open, displaying the following configuration:

- Branch:** production (Default)
- Compute:** Primary (Active)
- Database:** neondb
- Role:** neondb_owner (with a "Reset password" link)
- Connection pooling:** Enabled (toggle switch)

Below the configuration, there is a "Create new database" section with a list containing "neondb" and "mobile_db". At the bottom of the modal, it states: "Your password is saved in a secure storage vault. [Learn more about connecting](#)".

The background interface shows the "Project dashboard" for "solitary-hill-79473356?database=neondb". The left sidebar includes navigation options like "Dashboard", "Branches", "Integrations", "Settings", "Overview", "Monitoring", "SQL Editor", "Tables", "Backup & Restore", "Data Masking (BETA)", "Data API", and "Auth". The right sidebar shows project details like "Region: AWS US East 1 (N. Virginia)" and "Default compute size: .25 ↔ 2 CU".

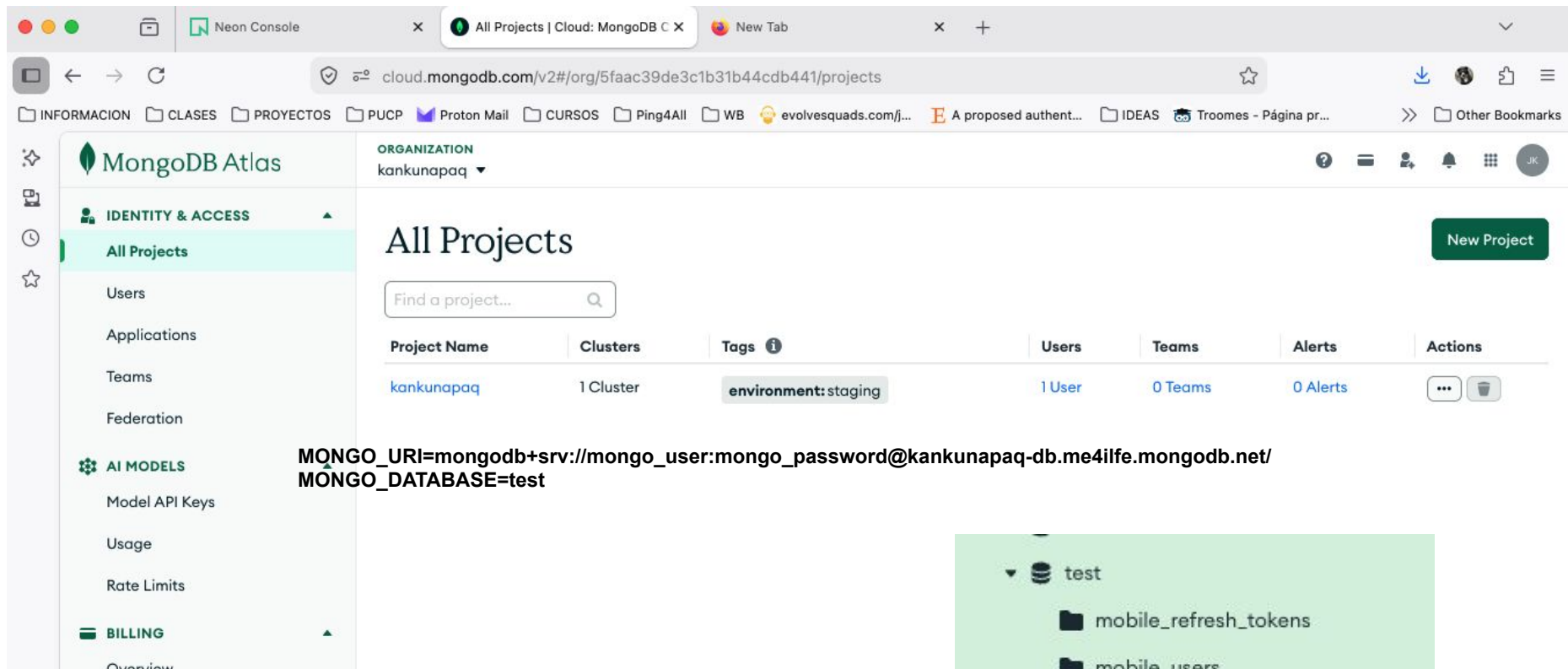
POSTGRES_HOST=localhost
POSTGRES_PORT=5432
POSTGRES_DB=mobile_db
POSTGRES_USER=postgres
POSTGRES_PASSWORD=postgres



```
CREATE TABLE IF NOT EXISTS geo_events (
  id BIGSERIAL PRIMARY KEY,
  user_id UUID,
  latitude DOUBLE PRECISION NOT NULL,
  longitude DOUBLE PRECISION NOT NULL,
  altitude DOUBLE PRECISION,
  accuracy DOUBLE PRECISION,
  speed DOUBLE PRECISION,
  heading DOUBLE PRECISION,
  event_type TEXT NOT NULL DEFAULT 'gps_ping',
  device_id TEXT,
  platform TEXT,
  app_version TEXT,
  device_model TEXT,
  recorded_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
  created_at TIMESTAMPTZ NOT NULL DEFAULT NOW()
);

CREATE TABLE IF NOT EXISTS files (
  id BIGSERIAL PRIMARY KEY,
  project_slug TEXT NOT NULL,
  user_id TEXT NOT NULL,
  storage_provider TEXT NOT NULL,
  bucket TEXT NOT NULL,
  object_key TEXT NOT NULL,
  url TEXT,
  file_name TEXT NOT NULL,
  content_type VARCHAR,
  size_bytes BIGINT,
  is_public BOOLEAN NOT NULL DEFAULT FALSE,
  uploaded_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
  deleted_at TIMESTAMPTZ
);

select * from files;
```

MongoDB Atlas

ORGANIZATION
kankunapaq

All Projects

Find a project...

Project Name	Clusters	Tags	Users	Teams	Alerts	Actions
kankunapaq	1 Cluster	environment: staging	1 User	0 Teams	0 Alerts	...

MONGO_URI=mongodb+srv://mongo_user:mongo_password@kankunapaq-db.me4ilfe.mongodb.net/
MONGO_DATABASE=test

```
test
├── mobile_refresh_tokens
├── mobile_users
└── score_searchins
```

For Startups: Get up to \$15,000 in Atlas credits for this and future clusters through MongoDB for Startups. [Apply today](#) ➔

DATABASE ACCESS

Database Users

Custom Roles

NETWORK ACCESS

IP Access List

Peering

Private Endpoint

Advanced

IP Access List

You will only be able to connect to your cluster from the following list of IP Addresses:

IP Address
0.0.0.0/0 (includes your...
179.619.242/32 (includes...

+ADD IP ADDRESS

Status	Actions
● Active	EDIT DELETE
● Active	EDIT DELETE

×

Add IP Access List Entry

Atlas only allows client connections to a cluster from entries in the project's IP Access List. Each entry should either be a single IP address or a CIDR-notated range of addresses. [Learn more](#)

Access List Entry:

15.235.49.139

Comment:

Servidor del Curso

This entry is temporary and will be deleted in

6 hours

Cancel

Confirm

Alert System Status: All Good

©2026 MongoDB, Inc. [Status](#) [Terms](#) [Privacy](#) [Atlas Blog](#) [Contact Sales](#)

← → ↺

dash.cloudflare.com/cbf3c2d1678949e1ec56424f2eaaccb8/r2/overview

🔖 ⭐ 📄 ⬇️ 🌑 ⋮

 Johnkbarrera@gmail.co...

⚡ Ask AI 🛡️ Support 👤

🔍 Quick search... 🔑

🏠 Account home

🕒 Recents >

🌐 Domains >

Observe

📁 Investigate >

🕒 Analytics >

Build

⚙️ Compute >

🧠 AI >

Storage & databases ▾

R2 Object Storage ▾

Overview

Data migration

R2 Data Catalog Beta

Postgres & MySQL (Hyperd...

Workers KV

D1 SQLite Database

Analytics Engine

Pipelines Beta >

Secrets Store New

📺 Media >

Protect & Connect

 **R2 Object Storage**

High-performance storage for files and objects with zero egress charges.

🔍 Search buckets...

↺

Buckets	Objects	Size
abc	--	--
kankunapag	3	767.3 kB
mobile	2	384.3 kB

<

>

Documentation + Create bucket

Usage

June 1 - July 1

Billing Dashboard →

Billable usage (current period) ⓘ + Add Budget Alert

\$0.00

No billable usage incurred yet

Class A Operations ⓘ

130

Class B Operations ⓘ

63

Total storage ⓘ

1.15 MB

Account Details

API Tokens {} Manage

Account ID cbf3c2d1678949e1ec56424f2ea... 📄

S3 API https://cbf3c2d1678949e1ec56424f2... 📄

- ☐ **Admin Read only:** Allows the ability to list buckets and view bucket configuration, read and list objects, and read access to data catalog tables and associated metadata.
- ☐ **Object Read & Write:** Allows the ability to read, write, and list objects in specific buckets.
- ☐ **Object Read only:** Allows the ability to read and list objects in specific buckets.

Permissions can be further customized or created for multiple accounts from your [Profile](#).

TTL

Define how long this token will stay active:

6 months ▼

Client IP Address Filtering

Select IP addresses or ranges of IP addresses to filter. This filter limits the client IP addresses that can use the API token with Cloudflare. By default, this token will apply to all addresses.

Include:

0.0.0.0 × 15.235.49.139 × e.g. 192.168.1.88

Invalid IP address or IP address range: http://15.235.49.139/

Exclude:

0.0.0.0 × 15.235.49.139 × e.g. 192.168.1.88

Cancel

Create User API Token

Create User API Token

✔ R2 User Token for Students was successfully created

Summary:

Permissions:	Allows the ability to create, list, and delete buckets, edit bucket configuration, read, write, and list objects, and read and write access to data catalog tables and associated metadata.
Buckets:	All R2 buckets on this account
TTL:	Active from Jun 6, 2026 - Dec 6, 2026

Use this token for authenticating against the [Cloudflare API](#):

Token value

cfut_nZH1TWVH0u0tc4goekTgD8JGvJE9LyXFmRSOTpV0a64fdbf9

Click to copy

OBJECT STORAGE BUCKET=mobile

OBJECT STORAGE ENDPOINT=https://cbf3c2d1678949e1ec56424f2eaaccb8.cloudflarestorage.com

Use the following credentials for S3 client

OBJECT STORAGE ACCESS KEY=8548d4dad51bc9b52f0bd7aefff1767

Access Key ID

8548d4dad51bc9b52f0bd7aefff1767

Click to copy

OBJECT STORAGE SECRET KEY=eb89427d67ef8072d76516d6b2208f4c257b703eb714547215261202963fb0f9

Secret Access Key

eb89427d67ef8072d76516d6b2208f4c257b703eb714547215261202963fb0f9

Click to copy

OBJECT STORAGE REGION=auto

OBJECT STORAGE TOKEN VALUE=cfut_nZH1TWVH0u0tc4goekTgD8JGvJE9LyXFmRSOTpV0a64fdbf9

Use jurisdiction-specific endpoints for S3 clients:

Default European Union (EU)

https://cbf3c2d1678949e1ec56424f2eaaccb8.r2.cloudflarestorage.com

Click to copy

For security reasons, these token values will not be shown again. [Learn more about API token generation](#)

Finish

python

```
import boto3

# Replace these values with your actual Cloudflare credentials
account_id = "your_cloudflare_account_id_here"
access_key_id = "your_access_key_id_here"
secret_access_key = "your_secret_access_key_here"

# Build the custom R2 S3 endpoint URL
r2_endpoint = f"https://{account_id}.r2.cloudflarestorage.com"

# Create the client
r2_client = boto3.client(
    service_name="s3",
    endpoint_url=r2_endpoint,
    aws_access_key_id=access_key_id,
    aws_secret_access_key=secret_access_key
)
```

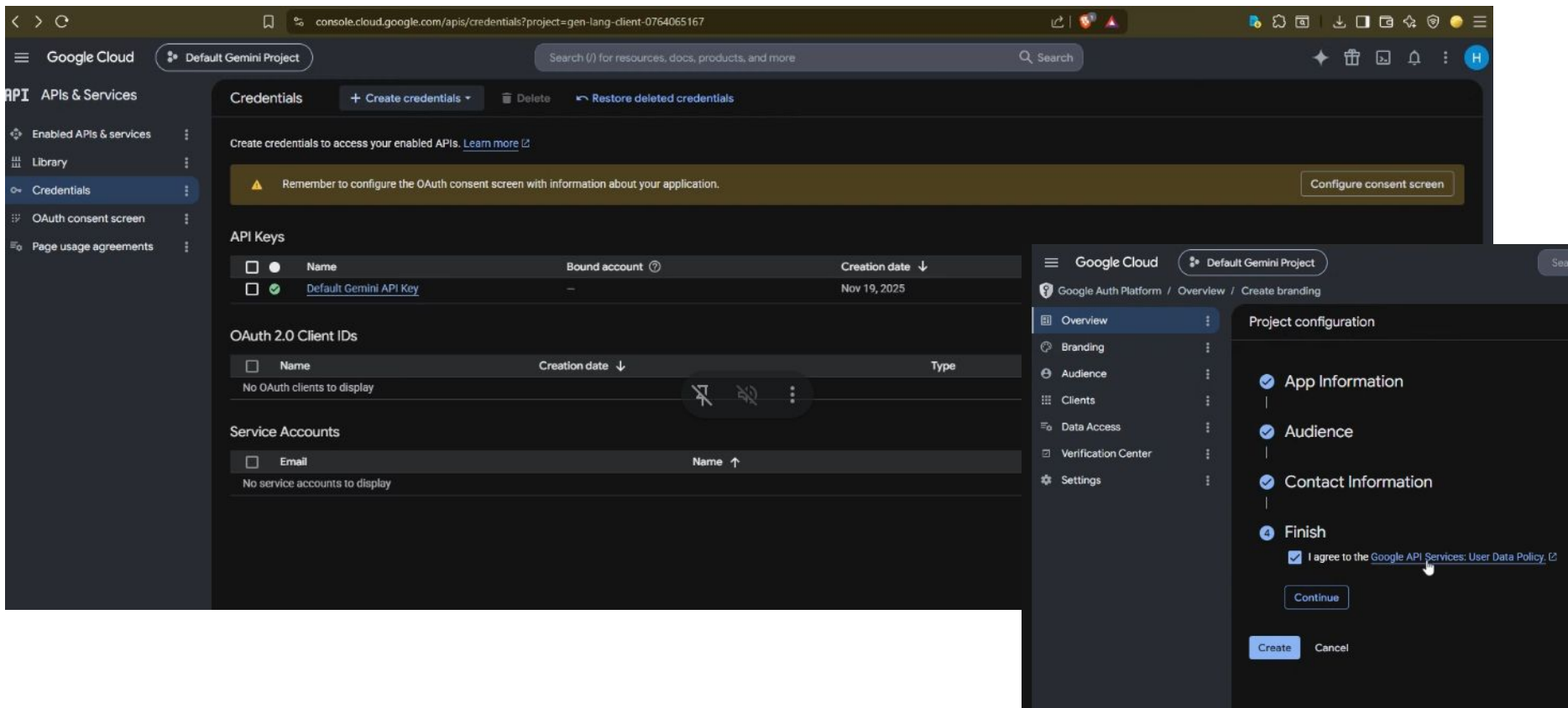
python

```
bucket_name = "my-r2-bucket"
local_file_path = "document.pdf"
r2_object_name = "uploads/document.pdf"

r2_client.upload_file(
    Filename=local_file_path,
    Bucket=bucket_name,
    Key=r2_object_name
)

print("Upload successful")
```

Credential Google



The screenshot displays the Google Cloud Console interface for the 'Default Gemini Project'. The left sidebar shows the navigation menu with 'APIs & Services' selected. The main content area is divided into two sections: 'Credentials' and 'OAuth 2.0 Client IDs'.

Credentials Section:

- Buttons: '+ Create credentials', 'Delete', 'Restore deleted credentials'.
- Text: 'Create credentials to access your enabled APIs. [Learn more](#)'.
- Warning: 'Remember to configure the OAuth consent screen with information about your application. [Configure consent screen](#)'.
- API Keys Table:**

☐	Name	Bound account	Creation date
☒	Default Gemini API Key	—	Nov 19, 2025

OAuth 2.0 Client IDs Section:

- Buttons: '+ Create OAuth 2.0 client ID', 'Delete', 'Restore deleted client IDs'.
- Text: 'No OAuth clients to display'.
- Buttons: '+ Create OAuth 2.0 service account', 'Delete', 'Restore deleted service accounts'.
- Text: 'No service accounts to display'.

Right Panel: Project configuration

- Navigation: Overview, Branding, Audience, Clients, Data Access, Verification Center, Settings.
- Steps: 1. App Information, 2. Audience, 3. Contact Information, 4. Finish.
- Agreement: ☒ I agree to the [Google API Services: User Data Policy](#).
- Buttons: 'Continue', 'Create', 'Cancel'.

Credential Google

← Create OAuth client ID

A client ID is used to identify a single app to Google's OAuth servers. If your app runs on multiple platforms, each will need its own client ID. See [Setting up OAuth 2.0](#) for more information. [Learn more](#) about OAuth client types.

Application type *
Desktop app ▼

Name *
Mobile endpoints

The name of your OAuth 2.0 client. This name is only used to identify the client in the console and will not be shown to end users.

Note: It may take 5 minutes to a few hours for settings to take effect

Create Cancel

OAuth client created

The client ID can always be accessed from Clients tab under Google Auth Platform.

ⓘ OAuth access is restricted to the [test users](#) listed on your [OAuth consent screen](#)

Client ID 5375...eb2qub.apps...usercontent.com

⚠ You will no longer be able to view or download the client secret once you close this dialog. Make sure you have copied or downloaded the information below and securely stored it.

Client secret G...NVvjdt9itoAERhR

Creation date June 6, 2026, 12:02:13 PM GMT-5

Status ✓ Enabled

⬇ Download JSON

OK

INFORMACIONCLASESPROYECTOSPUCPProton MailCURSOSPing4AllWBevolvesquads.com/jf...A proposed authent...IDEASTroomes - Página pr...ChatbotOther Bookmarks

illarek-lab / platform-api

Type / to search

<> CodeIssuesPull requestsAgentsActionsProjectsWikiSecurity and qualityInsightsSettings

platform-apiPublic

Edit PinsWatch 0Fork 0Star 0

main3 Branches0 TagsGo to fileAdd fileCode

johnkbarrera Merge pull request #5 from illarek-lab/feature-device-id-and-ref... 74c4a51 · 2 days ago 60 Commits

.github	feat: Add CODEOWNERS file to require approval from repo...	3 days ago
credentials	refactor: replace bcrypt with PBKDF2 for password hashin...	2 days ago
docs	docs: add Google auth testing gdocs: update Postman col...	3 days ago
src/app	refactor: replace bcrypt with PBKDF2 for password hashin...	2 days ago
.gitignore	Add initial project structure with dependencies and docum...	5 days ago
LICENSE	Initial commit	5 days ago
README.md	feat: Update README with project details and local setup i...	3 days ago
pyproject.toml	refactor: replace bcrypt with PBKDF2 for password hashin...	2 days ago
uv.lock	refactor: replace bcrypt with PBKDF2 for password hashin...	2 days ago

READMEAGPL-3.0 license

Illarek Lab — Platform API

A multi-tenant FastAPI backend platform that hosts independent projects under a single application. Each project exposes its own endpoints, maintains isolated infrastructure, and follows a clean layered architecture.

Overview

About

Platform API es una plataforma backend construida con FastAPI que permite alojar múltiples proyectos independientes dentro de una única aplicación.

- Readme
- AGPL-3.0 license
- Activity
- Custom properties
- 0 stars
- 0 watching
- 0 forks
- Audit log
- Report repository

Releases

No releases published
[Create a new release](#)

Packages

No packages published
[Publish your first package](#)

Contributors 2

- johnkbarrera John Barrera
- illareklab-code

People

Find a member...

Export

Invite member

Organization permissions

Members

11

Outside collaborators

Pending collaborators

Invitations

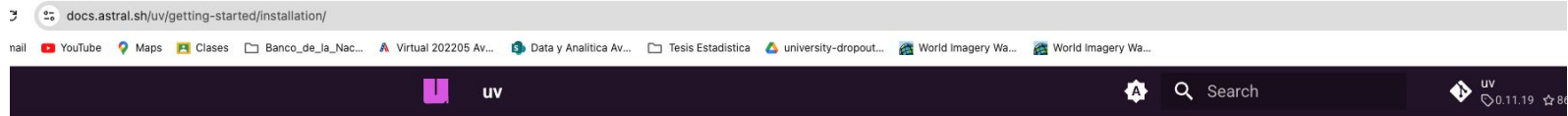
8

Failed invitations

Security Managers

You are the only owner of this organization! We recommend a minimum of two people within each organization have the owner role.

☐	Members	Two-factor authentication	Membership
☐	 Angelblazel Membership via: direct assignment	2FA Private Member 1 team 0 roles	...
☐	 Emil Helmut Zapata Arellano EmilZapata Membership via: direct assignment	2FA Private Member 1 team 0 roles	...
☐	 Fabrisio Aliaga FabriAlitT Membership via: direct assignment	2FA Private Member 1 team 0 roles	...
☐	 Frank Cristian Prado Ccopa frankhdprado2-maker Membership via: direct assignment	2FA Private Member 1 team 0 roles	...
☐	 Héctor David Martínez Huamán HectorD28 Membership via: direct assignment	2FA Private Member 1 team 0 roles	...



Introduction

Getting started

Installation

First steps

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Getting help

Guides

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Working on projects

Publishing packages

Migration

Integrations

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Tools

Python versions

Configuration files

Package indexes

Resolution

Build backend

Authentication

Caching

Preview features

Getting started

Installing uv

Installation methods

Install uv with our standalone installers or your package manager of choice.

Standalone installer

uv provides a standalone installer to download and install uv:

macOS and Linux Windows

Use `curl` to download the script and execute it with `sh`:

```
$ curl -LsSf https://astral.sh/uv/install.sh | sh
```

If your system doesn't have `curl`, you can use `wget`:

```
$ wget -qO- https://astral.sh/uv/install.sh | sh
```

Request a specific version by including it in the URL:

```
$ curl -LsSf https://astral.sh/uv/0.11.19/install.sh | sh
```

Installation methods

Standalone installers

PyPI

Homebrew

MacPorts

WinGet

Scoop

Docker

GitHub Releases

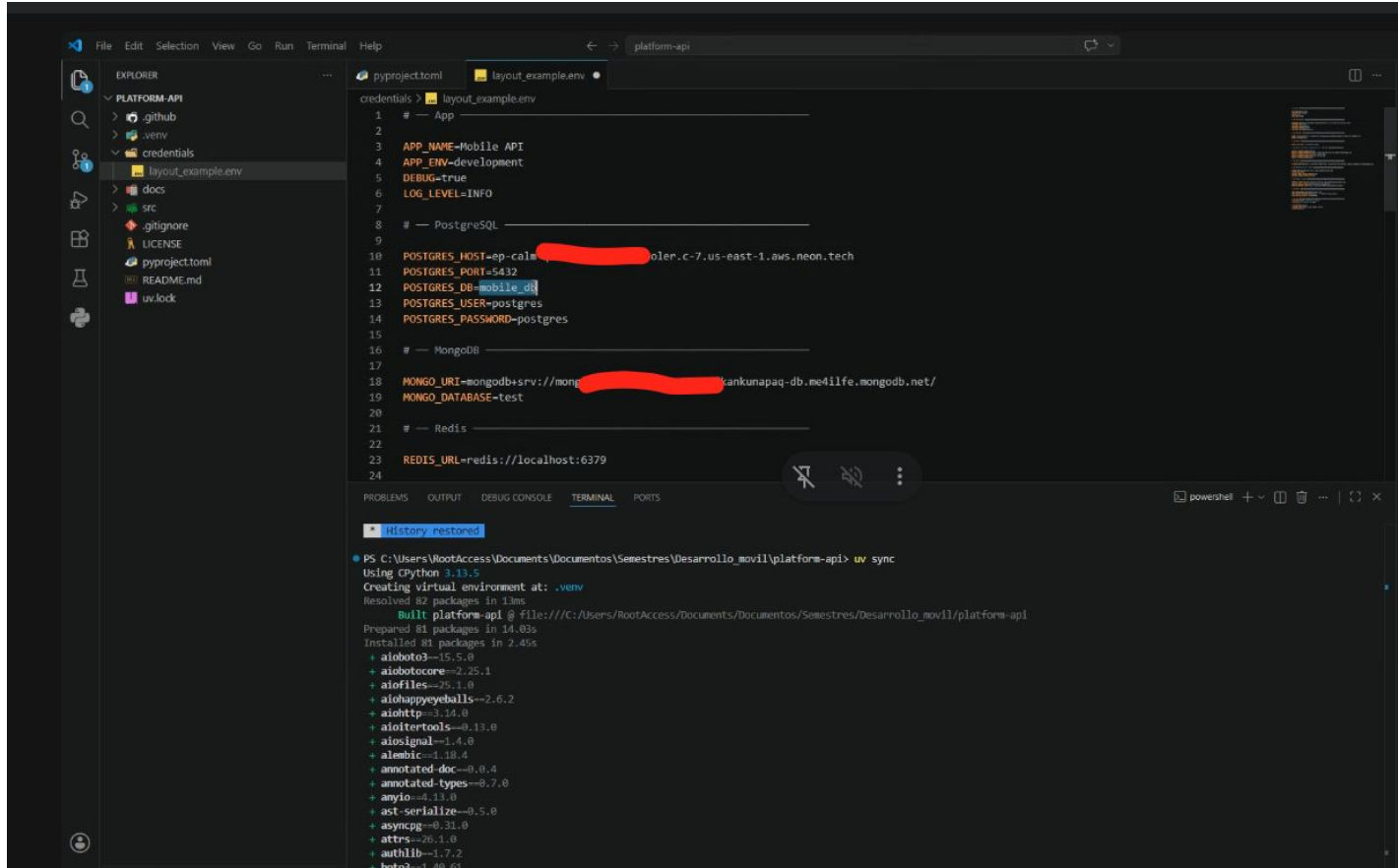
Cargo

Upgrading uv

Shell autocompletion

Uninstallation

Next steps



The screenshot shows a Visual Studio Code editor with a project named 'platform-api'. The Explorer sidebar on the left shows the file structure, including a 'credentials' folder containing 'layout_example.env'. The main editor area displays the contents of 'layout_example.env', which is a configuration file for a Django application. The file contains settings for the application name, environment, debug mode, logging, and database connections (PostgreSQL, MongoDB, and Redis). Some sensitive information, such as database hostnames and passwords, is redacted with black boxes.

```
1 # -- App
2
3 APP_NAME=Mobile API
4 APP_ENV=development
5 DEBUG=true
6 LOG_LEVEL=INFO
7
8 # -- PostgreSQL
9
10 POSTGRES_HOST=ep-calc-1.us-east-1.aws.neon.tech
11 POSTGRES_PORT=5432
12 POSTGRES_DB=mobile_db
13 POSTGRES_USER=postgres
14 POSTGRES_PASSWORD=postgres
15
16 # -- MongoDB
17
18 MONGO_URI=mongodb+srv://mongobackup:Kankunapaq-db.me4ilife.mongodb.net/
19 MONGO_DATABASE=test
20
21 # -- Redis
22
23 REDIS_URL=redis://localhost:6379
24
```

Below the editor, the TERMINAL panel is open, showing the output of a command executed in a PowerShell terminal. The command was 'uv sync' in the directory 'C:\Users\RootAccess\Documents\Semestres\Desarrollo_movil\platform-api'. The output shows that a virtual environment was created and packages were installed.

```
PS C:\Users\RootAccess\Documents\Semestres\Desarrollo_movil\platform-api> uv sync
Using CPython 3.13.5
Creating virtual environment at: .venv
Resolved 82 packages in 13ms
Built platform-api @ file:///C:/Users/RootAccess/Documents/Semestres/Desarrollo_movil/platform-api
Prepared 81 packages in 14.03s
Installed 81 packages in 2.45s
+ aiohttp==3.14.0
+ aiohttp-jinja2==1.4.0
+ aiohttp-multipart==0.10.0
+ aiohttp-retry==2.8.3
+ aiohttp-xml==0.1.0
+ aiohttp-yaml==0.1.0
+ aiohttp-zlib-ng==0.3.0
+ aiohttp==3.14.0
+ aioitertools==0.13.0
+ aiosignal==1.4.0
+ alembic==1.18.4
+ annotated-doc==0.0.4
+ annotated-types==0.7.0
+ anyio==4.13.0
+ ast-serializable==0.5.0
+ asynccpg==0.31.0
+ attrs==26.1.0
+ authlib==1.7.2
+ boto3==1.40.61
```

People

Find a member...

Export

Invite member

Organization permissions

Members 11

Outside collaborators

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Failed invitations

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Gracias

